

Optimization Methods

Seminar 01**Optimization as the Foundation of Machine Learning**

Theory handout

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This handout collects the theory behind Seminar 1: what an optimization problem is, derivatives of functions of one and several variables, Taylor's formula, definiteness of matrices, optimality conditions and convexity. It is written for reading from zero: a derivative from a first calculus course is all you need, and vectors, determinants, gradients, Hessians and eigenvalues are introduced on the way. Every statement comes with an example with numbers; the main proofs start with their idea in one sentence; two kinds of boxes mark the traps where almost everyone slips and short questions to check yourself. The section on seminar problems contains all eleven problems with complete solutions; the home practice, common mistakes and a glossary follow.

Notation and conventions.

Vectors are columns: $x = (x_1, \dots, x_n)^\top \in \mathbb{R}^n$, and $x^\top y = x_1 y_1 + \dots + x_n y_n$ is a number. $\|x\| = \sqrt{x^\top x}$ is the Euclidean length. Every problem is a minimization, $\min_{x \in \Omega} f(x)$: a maximum of f is a minimum of $-f$. The gradient $\nabla f(x)$ is a column vector and $H(x)$ is the Hessian; for every function in this handout H is symmetric. x^* denotes a candidate point or a minimizer. $A \succ 0$ and $A \succeq 0$ mean that a symmetric matrix A is positive definite and positive semidefinite. A step along a direction s is $x + \alpha s$ with a step parameter $\alpha \geq 0$; it is a length only when $\|s\| = 1$. The letter λ is a weight in $[0, 1]$ or an eigenvalue, and the context always tells which.

1 Notation

Formulas are easier to read when you can say them aloud. The table lists the symbols, how to read them and what they mean.

Symbol	How to read it	Meaning
$\mathbb{R}^n$	"R n"	ordered lists of n real numbers, for example $x = (1, -2, 0.5) \in \mathbb{R}^3$
$\ x\ $	"the norm of x"	the length $\sqrt{x_1^2 + \dots + x_n^2}$
$x^\top y$	"x transpose y", "the inner product"	$x^\top$ is the column x written as a row; $x^\top y = x_1 y_1 + \dots + x_n y_n$ is a single number
∇f	"nabla f", "grad f"	the column of first partial derivatives
$\partial f / \partial x_1$	"partial f partial x one"	the partial derivative with respect to x_1
H	"the Hessian", "H of x"	the matrix of second partial derivatives
Ω	"omega"	the feasible set, where the minimum is sought
x^*	"x star"	a candidate point or a minimizer
$A \succ 0, A \succeq 0$	"A is positive definite", "positive semidefinite"	$h^\top A h > 0$ for all $h \neq 0$ (all eigenvalues positive); $h^\top A h \geq 0$ for all h (all eigenvalues nonnegative)
$o(\alpha)$	"little o of alpha"	a quantity with $o(\alpha)/\alpha \rightarrow 0$ as $\alpha \rightarrow 0$

The inner product is a number

$x = (1, -1)$ and $y = (2, 4)$ give $x^\top y = 1 \cdot 2 + (-1) \cdot 4 = -2$: a number, not a vector. In the section on optimality conditions this number becomes the rate of change of a function along a direction.

2 Optimization problems

The whole seminar tells one story. We want the points where a function takes its smallest value. Checking every point is impossible, so we learn to recognize a minimum by local signs, by the derivatives at the point itself: the first derivative produces candidates, the second one sorts them and in rare cases only a direct argument helps.

2.1 What we minimize and why

An optimization problem is written as

$$\min_{x \in \Omega} f(x).$$

Here f is the **objective function**, $x \in \mathbb{R}^n$ are the **variables** and Ω is the **feasible set**, the points among which the minimum is sought. With two variables a candidate is a point of the plane, with three a point of space. With many variables there is no picture, but the formulas are the same.

In machine learning the variables are the parameters of a model, and the function is its error on the data. A linear model with 50 weights is a point of $\mathbb{R}^{50}$, a neural network has about 10^9 parameters. With training data (a_i, y_i) , $i = 1, \dots, m$, a feature vector a_i and a target y_i , **training** means minimizing the **empirical risk**

$$L(w) = \frac{1}{m} \sum_{i=1}^m \ell(y_i, w^\top a_i),$$

where the loss ℓ measures the error on one object, for example $\ell(y, \hat{y}) = (y - \hat{y})^2$.

Example. Fit a line $y = w_1 + w_2 t$ to the points (t_i, y_i) with $t = (0, 1, 2)$ and $y = (1, 2, 4)$. With the squared loss, training is the problem

$$\min_{w \in \mathbb{R}^2} L(w), \quad L(w) = \frac{1}{3}((w_1 - 1)^2 + (w_1 + w_2 - 2)^2 + (w_1 + 2w_2 - 4)^2).$$

The answer $w^* = (\frac{5}{6}, \frac{3}{2})$ with $L(w^*) = \frac{1}{18}$ is derived in the section on convexity, once gradients and convexity are available.

Why. Brute force is hopeless: a grid with only 10 values per coordinate has 10^{50} points in $\mathbb{R}^{50}$. Instead of searching everywhere we look at one point and ask what its derivatives say.

2.2 The feasible set

The feasible set $\Omega \subseteq \mathbb{R}^n$ collects the points that are allowed. When $\Omega = \mathbb{R}^n$, the problem is **unconstrained**. Typical feasible sets are the quadrant $\{x_1 \geq 0, x_2 \geq 0\}$, the disk $\|x\| \leq 2$ and, in general, a set described by a system of inequalities.

Definition: Interior and boundary points. A point $x \in \Omega$ is **interior** if some ball $\{y : \|y - x\| < \varepsilon\}$ with $\varepsilon > 0$ lies in Ω ; otherwise it is a **boundary** point.

Example. In the quadrant $\Omega = \{x_1 \geq 0, x_2 \geq 0\}$ the point $(3, 5)$ is interior. The point $(0, 2)$ is a boundary point: a step to the left, however small, leaves Ω . The corner $(0, 0)$ is a boundary point too. This distinction becomes decisive in the section on optimality conditions.

2.3 Local and global minima

Definition: Local minimum. A point $x^* \in \Omega$ is a **local minimum** of f on Ω if there is $\varepsilon > 0$ such that

$$f(x) \geq f(x^*) \quad \text{for all } x \in \Omega \text{ with } \|x - x^*\| < \varepsilon.$$

Read the definition clause by clause. “There is $\varepsilon > 0$ ”: some radius exists, possibly a very small one. “For all $x \in \Omega$ with $\|x - x^*\| < \varepsilon$ ”: x^* is compared only with feasible points closer than this

radius. “ $f(x) \geq f(x^*)$ ”: none of them is lower, and equal values are allowed. In short, x^* is a lowest feasible point in some neighbourhood of itself.

Definition: Global minimum. A point $x^* \in \Omega$ is a **global minimum** of f on Ω if $f(x) \geq f(x^*)$ for all $x \in \Omega$, with no restriction on the distance.

Definition: Strict minimum. A minimum is **strict** if $f(x) > f(x^*)$ for every admissible $x \neq x^*$: for all $x \in \Omega$ with $0 < \|x - x^*\| < \varepsilon$ in a **strict local minimum** and for all $x \in \Omega$ with $x \neq x^*$ in a **strict global minimum**.

Example. $f(x) = x_1^2$ on $\mathbb{R}^2$ satisfies $f \geq 0$ everywhere and $f = 0$ on the whole line $x_1 = 0$. Every point of this line is a global minimum, and none of them is strict, because each has neighbours on the line with the same value. Such a line is a flat bottom.

Local and global

A global minimum is a local one whose neighbourhood is all of Ω . Derivatives are local tools; a global claim needs convexity, a direct bound or, when a global minimum is known to exist (for example when $f(x) \rightarrow +\infty$ as $\|x\| \rightarrow \infty$, or when Ω is closed and bounded), a comparison of all local minima.

Check yourself. A function on the line tends to $+\infty$ as $|x| \rightarrow \infty$ and has two wells of different depth. How many local and how many global minima does it have?

Answer. Two local minima and one global, at the bottom of the deeper well; a global minimum exists because f grows at infinity. If the two wells have equal depth, both bottoms are global minima; each is still a strict local minimum, but neither is a strict global minimum.

3 Warm-up: one variable

We start with the derivative from a first calculus course and look at it with the eyes of optimization.

3.1 The derivative is a rate

The **derivative** of $f : \mathbb{R} \rightarrow \mathbb{R}$ at a point x is the limit

$$f'(x) = \lim_{\alpha \rightarrow 0} \frac{f(x + \alpha) - f(x)}{\alpha}.$$

Take a small step α and divide the change of f by the step: the result is a rate, the change of f per unit of step. Geometrically, $f'(x)$ is the slope of the tangent to the graph at x . A one-sided version, with $\alpha \rightarrow 0^+$, appears with directional derivatives.

If $f'(x) \neq 0$ at a point inside the domain, then f decreases on one side of x , so x is not a minimum.

3.2 A zero derivative is necessary, not sufficient

Example. $f(x) = x^3$ has $f'(x) = 3x^2$ and $f'(0) = 0$, yet $f < 0$ to the left of 0 and $f > 0$ to the right: 0 is no extremum.

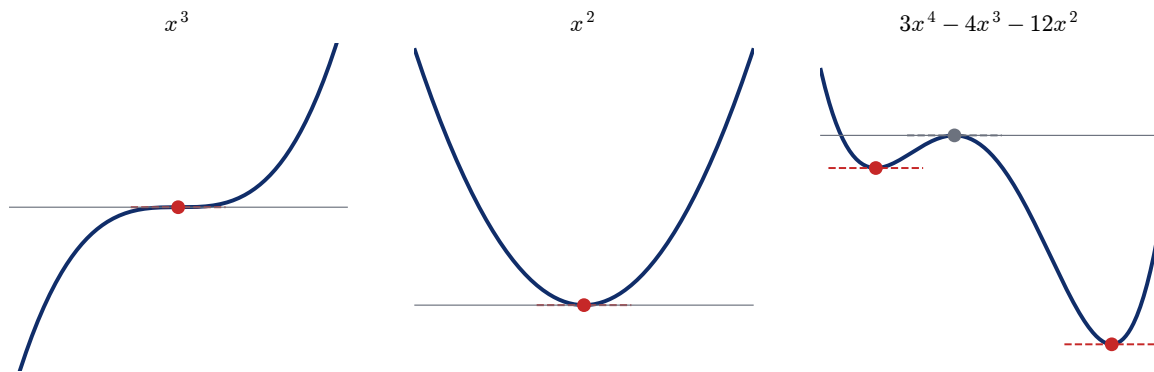


Figure. Left: x^3 has $f'(0) = 0$ but no extremum. Middle: x^2 has a strict minimum at 0. Right: $f(x) = 3x^4 - 4x^3 - 12x^2$ with $f'(x) = 12x(x+1)(x-2)$ has local minima at $x = -1$ and $x = 2$ and a local maximum at $x = 0$; only the deeper well, $f(2) = -32$, is the global minimum.

Necessary and sufficient

A condition is **necessary** if it holds at every minimum, and **sufficient** if it guarantees a minimum. At interior points $f'(x) = 0$ is necessary but not sufficient: solving $f' = 0$ gives candidates, not answers.

The second derivative tells x^2 from x^3 : for x^2 it gives $f''(0) = 2 > 0$, while for x^3 it gives $f''(0) = 0$, which carries no information.

3.3 Taylor's formula

Taylor's formula replaces a function near a point by a polynomial. For a twice differentiable f ,

$$f(x + \alpha) = \underbrace{f(x)}_{\text{value}} + \underbrace{f'(x)\alpha}_{\text{linear part}} + \underbrace{\frac{1}{2}f''(x)\alpha^2}_{\text{curvature}} + o(\alpha^2).$$

The remainder $o(\alpha^2)$ satisfies $o(\alpha^2)/\alpha^2 \rightarrow 0$ as $\alpha \rightarrow 0$: for small steps it is negligible next to the other terms.

How to use it

If $f'(x) \neq 0$, the linear term decides: a small step with $f'(x)\alpha < 0$ decreases f , so x is not a minimum. If $f'(x) = 0$, the sign of $f''(x)$ decides: $f''(x) > 0$ gives a strict local minimum and $f''(x) < 0$ a strict local maximum. If also $f''(x) = 0$, the formula is silent. The section on Taylor's formula in several variables repeats this logic for many variables.

Example. For $f(x) = 3x^4 - 4x^3 - 12x^2$ we have $f'(x) = 12x(x+1)(x-2)$ and $f''(x) = 36x^2 - 24x - 24$. At the stationary points: $f''(-1) = 36 > 0$, a local minimum; $f''(0) = -24 < 0$, a local maximum; $f''(2) = 72 > 0$, a local minimum. Which of the two minima is global the derivatives do not say. Since $f(x) \rightarrow +\infty$ as $|x| \rightarrow \infty$, a global minimum exists and is one of the local minima, so compare the values $f(-1) = -5$ and $f(2) = -32$.

Check yourself. Let $f(x) = x^4$. Find $f'(0)$ and $f''(0)$. Is 0 a minimum?

Answer. Both are zero, so the test is silent. But $x^4 \geq 0$ for every x with equality only at 0, so 0 is a strict global minimum. The same situation returns in two variables in problem B4.

4 Functions of several variables

A function of n variables assigns a number to every point of $\mathbb{R}^n$. The examples of this section use the function from the lecture, $f = 3x_1^3 + 2x_1x_2^2 + 4x_2^2$, at the point $(1, 1)$; problem A2 evaluates the same function at another point.

4.1 Level sets: a height map

A function of two variables is easy to picture as a relief: $f(x_1, x_2)$ is the height above the point (x_1, x_2) . The **level set** $\{x : f(x) = c\}$ collects the points of equal height c , exactly like a contour line on a topographic map. Several level sets drawn with their values give a height map of f . Two different level sets never cross, because a point has only one height.

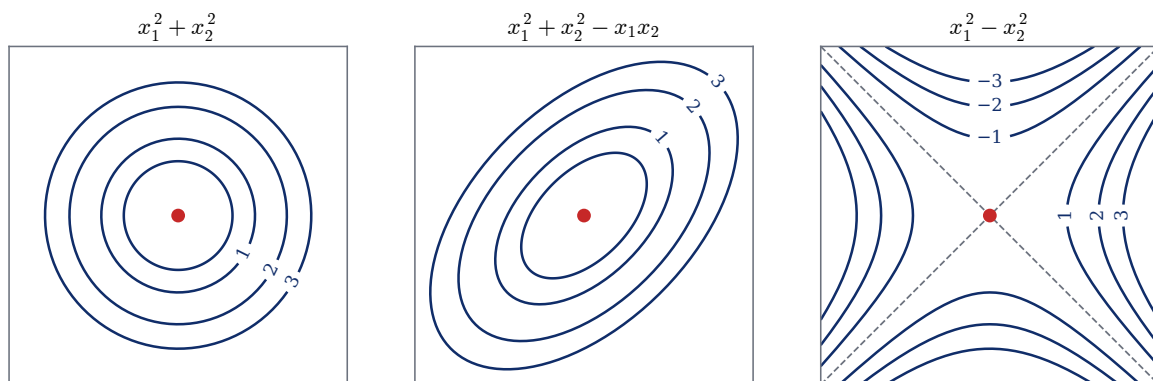


Figure. Level sets with their values. Left: $x_1^2 + x_2^2$ (problem B1), circles around a minimum. Middle: $x_1^2 + x_2^2 - x_1x_2$ (problem A1), tilted ellipses around a minimum. Right: $x_1^2 - x_2^2$ (problem B2), hyperbolas around a saddle; the dashed lines are the level 0.

Closed nested level sets surround a pit or a hill, and only the level values tell which: without them a pit and a hill look the same. Open level sets that turn away from each other mark a saddle: the function goes up in one direction and down in another. Where level sets with evenly spaced values crowd together, the function changes fast.

4.2 Partial derivatives

Definition: Partial derivative. The **partial derivative** of f with respect to x_i is the ordinary derivative in x_i with all other variables held fixed; it is the rate of change of f when only x_i moves:

$$\frac{\partial f}{\partial x_i}(x) = \lim_{\alpha \rightarrow 0} \frac{f(x_1, \dots, x_i + \alpha, \dots, x_n) - f(x)}{\alpha}.$$

In practice: treat every letter except x_i as a number and differentiate by the usual rules. The result is again a function of all n variables.

Example. Differentiate $f = 3x_1^3 + 2x_1x_2^2 + 4x_2^2$ term by term. With respect to x_1 : $3x_1^3$ gives $9x_1^2$; in $2x_1x_2^2$ the factor x_2^2 is a “number”, so the term gives $2x_2^2$; $4x_2^2$ does not contain x_1 and gives 0. So $\partial f / \partial x_1 = 9x_1^2 + 2x_2^2$. Likewise, with respect to x_2 the first term gives 0 and the other two give $4x_1x_2$ and $8x_2$, so $\partial f / \partial x_2 = 4x_1x_2 + 8x_2$.

Check yourself. Find both partial derivatives of $f = x_1^2x_2 + 5x_2^3$.

Answer. $\partial f / \partial x_1 = 2x_1x_2$ and $\partial f / \partial x_2 = x_1^2 + 15x_2^2$.

4.3 The gradient

Definition: Gradient. The **gradient** of f at x collects all partial derivatives into one column vector:

$$\nabla f(x) = (\partial f / \partial x_1, \dots, \partial f / \partial x_n)^\top \in \mathbb{R}^n.$$

- **The gradient points in the direction of steepest ascent**, and $-\nabla f$ in the direction of steepest descent. Its length $\|\nabla f\|$ is the rate of change in that direction per unit of length (when $\nabla f(x) \neq 0$). This is why gradient descent steps against the gradient; the section on optimality conditions proves it with the directional derivative.
- **The gradient is perpendicular to level sets.** Along a level set f does not change, so its rate of change in that direction is zero, and the gradient has no component along the level set.

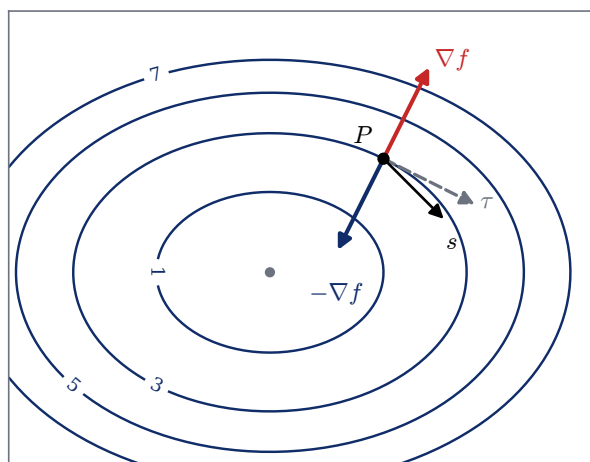


Figure. Level sets of $f = x_1^2 + 2x_2^2$ and the point $P = (1, 1)$ on the level $f = 3$. The gradient $\nabla f(P) = (2, 4)$ is perpendicular to this level set and points uphill; $-\nabla f(P)$ points downhill; along the tangent direction τ the rate of change is zero. The direction $s = (1, -1)$ of problem D2 makes an obtuse angle with $\nabla f(P)$, so f decreases along it.

Example. Evaluating the partial derivatives of the lecture function at the point gives $\nabla f(1, 1) = (9 + 2, 4 + 8)^\top = (11, 12)$. A small step α along $(11, 12)$ changes f by about $\alpha(121 + 144) = 265\alpha$; the opposite step decreases f by the same amount to first order. The factor in front of α is $\|\nabla f\|^2$, the inner product of the gradient with itself.

4.4 The Hessian

Differentiating a partial derivative again, in the same or in another variable, gives a second partial derivative. For n variables there are n^2 of them, and they are arranged in a square table.

Definition: Hessian. The **Hessian** of f at x is the $n \times n$ matrix of second partial derivatives; its entry in row i and column j is $\partial^2 f / \partial x_i \partial x_j$:

$$H(x) = [\partial^2 f / \partial x_i \partial x_j]_{i,j=1}^n.$$

Theorem: Schwarz. If the second partial derivatives of f are continuous, they do not depend on the order of differentiation: $\partial^2 f / \partial x_i \partial x_j = \partial^2 f / \partial x_j \partial x_i$. Hence the Hessian is symmetric.

All functions of the seminar are polynomials, whose second derivatives are continuous everywhere, so their Hessians are symmetric. This matters: everything later, eigenvalues and Sylvester's criterion included, is about symmetric matrices.

Example. The Hessian of the lecture function step by step. Differentiating $\partial f/\partial x_1 = 9x_1^2 + 2x_2^2$ in x_1 gives $18x_1$ and in x_2 gives $4x_2$; differentiating $\partial f/\partial x_2 = 4x_1x_2 + 8x_2$ in x_2 gives $4x_1 + 8$. So

$$H(x) = \begin{pmatrix} 18x_1 & 4x_2 \\ 4x_2 & 4x_1 + 8 \end{pmatrix}, \quad H(1, 1) = \begin{pmatrix} 18 & 4 \\ 4 & 12 \end{pmatrix},$$

and $\det H(1, 1) = 216 - 16 = 200$. Check the symmetry: differentiating $\partial f/\partial x_2$ in x_1 gives the same $4x_2$. Unlike the Hessian of a quadratic function, this one depends on the point, so problem A2 gets a different matrix at its point.

4.5 The cross-term trap

Trap. A term $c x_1 x_2$ gives $H_{12} = H_{21} = c$: differentiate directly, $\partial(c x_1 x_2)/\partial x_1 = c x_2$ and then $\partial(c x_2)/\partial x_2 = c$, with no halving. The value $c/2$ is the entry of A in $f = x^\top A x$: the product $x^\top A x$ produces $x_1 x_2$ twice, as $A_{12} x_1 x_2 + A_{21} x_2 x_1$, and $H = A + A^\top$ for a symmetric A . The form $f = \frac{1}{2} x^\top H x$ needs no halving. The two wrong values seen in practice are $c/2$ (the $x^\top A x$ convention) and $2c$ (doubling as if the term were a square).

Convention	Off-diagonal entry	For $f = x_1^2 + x_2^2 - x_1 x_2$
Hessian H	c	$H_{12} = -1$
$f = x^\top A x$	$c/2$	$A_{12} = -0.5$
$f = \frac{1}{2} x^\top H x$	c	the Hessian itself

The rule for reading H off a polynomial of degree two without differentiating: the diagonal holds twice the coefficient of each square; each cross term puts its coefficient into both symmetric positions; variables that never appear together give 0.

Check yourself. Write down the Hessian of $f = 3x_1^2 + x_2^2 + 2x_3^2 - 4x_1x_2 + 6x_2x_3$ without differentiating term by term.

Answer. The diagonal holds twice the coefficients of the squares, the off-diagonal entries hold the coefficients of the cross terms in both symmetric positions; x_1 and x_3 never meet:

$$H = \begin{pmatrix} 6 & -4 & 0 \\ -4 & 2 & 6 \\ 0 & 6 & 4 \end{pmatrix}.$$

5 Taylor's formula in several variables

Taylor's formula is the main tool of the course: near a point a smooth function is replaced by a polynomial with a small error, and every optimality condition is read off this formula.

5.1 The formula

For a twice continuously differentiable f and a small vector h , the value near x is the value at x plus a linear and a quadratic term, up to a small remainder:

$$f(x + h) = f(x) + \nabla f(x)^\top h + \frac{1}{2} h^\top H(x) h + o(\|h\|^2).$$

The remainder satisfies $o(\|h\|^2)/\|h\|^2 \rightarrow 0$ as $h \rightarrow 0$.

- $f(x)$: the value where we stand, the starting height.
- $\nabla f(x)^\top h$: the **linear part**, a single number, the inner product of the gradient with the shift. It changes sign with h : the steps h and $-h$ have linear parts of equal size and opposite signs.
- $\frac{1}{2}h^\top Hh$: the **quadratic part**, the curvature. It is unchanged under $h \rightarrow -h$, because h enters twice.

Why. If $\nabla f(x) \neq 0$, the linear part dominates for small steps, and one of h and $-h$ decreases f , so x is not a minimum. If $\nabla f(x) = 0$, the linear part vanishes and the sign of $h^\top Hh$ over all directions h decides; the section on quadratic forms and definiteness studies these signs.

5.2 Computing the quadratic term

The number $h^\top Hh$ is built from the entries of H and the components of h . For $n = 2$, multiplying out gives

$$h^\top Hh = H_{11}h_1^2 + 2H_{12}h_1h_2 + H_{22}h_2^2.$$

Such a sum of squares and products of the components with constant coefficients is called a **quadratic form**. The cross term appears twice, as $H_{12}h_1h_2$ and $H_{21}h_2h_1$, hence the factor 2: the same doubling as in the cross-term trap.

Example. For the lecture Hessian $H = \begin{pmatrix} 18 & 4 \\ 4 & 12 \end{pmatrix}$ and the step $h = (1, 1)$: $h^\top Hh = 38$ and $\frac{1}{2}h^\top Hh = 19$.

5.3 A numerical check

Numbers show that the formula really works. Take the lecture function at $(1, 1)$, where $f = 9$, $\nabla f = (11, 12)$ and $H = \begin{pmatrix} 18 & 4 \\ 4 & 12 \end{pmatrix}$. Along $h = t(1, 1)$ compare three numbers: the exact value, the linear model $9 + 23t$ and the quadratic model $9 + 23t + 19t^2$.

t	exact f	linear	quadratic	linear error	quadratic error
0.5	25.875	20.5	25.25	5.375	0.625
0.1	11.495	11.3	11.49	0.195	0.005
0.01	9.231905	9.23	9.2319	0.001905	0.000005

Dividing t by 10, as from the second row to the third, divides the linear error by about 100 (like t^2) and the quadratic error by 1000 (like t^3). Along this line f equals $5(1+t)^3 + 4(1+t)^2$, so the quadratic error is exactly $5t^3$. This is what $o(\|h\|^2)$ means: the remainder vanishes faster than the terms that are written out.

Quadratic functions: the formula is exact

A polynomial of degree two has no nonzero third derivatives, so the remainder is zero and Taylor's formula is exact. Check it on $q = x_1^2 + x_1x_2 + 2x_2^2$ at $(1, 2)$: $q = 11$, $\nabla q = (4, 9)$, $H = \begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}$. For $h = (0.3, -0.2)$ the linear part is -0.6 ; the products h_1^2 , h_1h_2 and h_2^2 , equal to $(0.09, -0.06, 0.04)$, give $h^\top Hh = 0.22$ and the quadratic part 0.11. The total is 10.51, equal to $q(1.3, 1.8)$. Linear regression with the squared loss is quadratic in the weights, so it is solved

by a formula, without iterations.

6 Quadratic forms and definiteness

Taylor's formula shows that at a point with $\nabla f(x) = 0$ the linear term vanishes and the quadratic term $\frac{1}{2} h^\top H(x) h$ takes over. Whether f goes up or down near x therefore depends on the signs that $h^\top H h$ takes as h ranges over all directions. This section studies which signs a quadratic form can take and how to find them.

6.1 Five cases

Definition: Definiteness. Let A be a symmetric $n \times n$ matrix. A is **positive definite**, $A \succ 0$, if $h^\top A h > 0$ for all $h \neq 0$, and **positive semidefinite**, $A \succeq 0$, if $h^\top A h \geq 0$ for all h . A is **negative definite**, $A \prec 0$, if $h^\top A h < 0$ for all $h \neq 0$, and **negative semidefinite**, $A \preceq 0$, if $h^\top A h \leq 0$ for all h . Finally, A is **indefinite** if $h^\top A h$ takes both signs: $h^\top A h > 0$ for some h and $g^\top A g < 0$ for some g .

In terms of a landscape, the graph of $h^\top A h$ is a bowl when $A \succ 0$, a dome when $A \prec 0$, a saddle when A is indefinite and a trough with a flat bottom when $A \succeq 0$ is singular: along the directions with $h^\top A h = 0$ the graph stays at height 0.

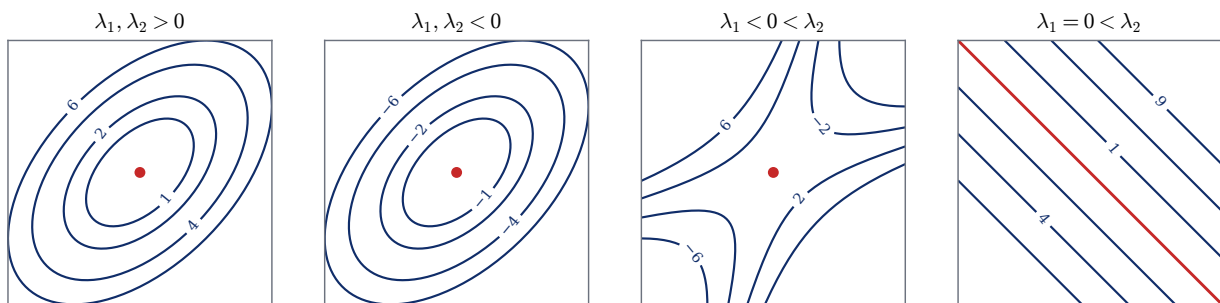


Figure. Level sets of $h^\top A h$: a bowl for $A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$, a dome for $A = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}$, a saddle for $A = \begin{pmatrix} 2 & -4 \\ -4 & 2 \end{pmatrix}$ and a trough for $A = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$, whose zero eigenvalue gives the flat bottom drawn in red.

6.2 Eigenvalues give the answer

A symmetric $n \times n$ matrix A has n mutually perpendicular unit **eigenvectors** $v_1, \dots, v_n$: along each of them A acts as a stretch, $Av_i = \lambda_i v_i$, and the stretch factors $\lambda_1, \dots, \lambda_n$ are the **eigenvalues**, all of them real. Write h in the axes of the eigenvectors, $h = u_1 v_1 + \dots + u_n v_n$. Then the cross terms disappear and the quadratic form becomes a weighted sum of squares:

$$h^\top A h = \lambda_1 u_1^2 + \dots + \lambda_n u_n^2.$$

Every question about the signs of $h^\top A h$ is now answered by the signs of the λ_i .

Eigenvalues	Type	Level sets
all > 0	positive definite	ellipses around a minimum
all < 0	negative definite	ellipses around a maximum
both signs	indefinite	hyperbolas, a saddle
all ≥ 0 , some zero	positive semidefinite	parallel lines, a trough
all ≤ 0 , some zero	negative semidefinite	the same, upside down

For a 2×2 matrix the eigenvalues are the roots of the quadratic equation $\det(A - \lambda I) = 0$, that is

$$\lambda^2 - (A_{11} + A_{22})\lambda + \det A = 0.$$

Example. $A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$: the equation is $\lambda^2 - 4\lambda + 3 = 0$ with the roots $(1, 3)$, both positive, so A is positive definite. $A = \begin{pmatrix} 2 & -4 \\ -4 & 2 \end{pmatrix}$: the equation is $\lambda^2 - 4\lambda - 12 = 0$ with the roots $(-2, 6)$ of opposite signs, so A is indefinite. These are the Hessians of problems A1 and C2.

The two matrices differ only in the size of the off-diagonal entry. So a cross term can destroy definiteness, and what decides is its size, not its sign: changing the sign of the off-diagonal entries of a 2×2 matrix does not change its eigenvalues. Problem C2 studies the whole family.

Looking ahead: the condition number

For $A \succ 0$ the ratio $\kappa = \lambda_{\max}/\lambda_{\min}$ is the **condition number**; it measures how stretched the level sets are. For $x_1^2 + x_2^2$ the eigenvalues are $(2, 2)$, so $\kappa = 1$ and the level sets are circles: a gradient step with step size $\frac{1}{2}$ lands exactly at the minimum, $x - \frac{1}{2} \cdot 2x = 0$. For the matrix of problem A1, $\kappa = 3$ and the level sets are ellipses. The larger κ , the narrower the ravine and the slower gradient methods: the later topics of the course return to this.

6.3 Sylvester's criterion

For a 3×3 matrix the eigenvalues are the roots of a cubic equation, which is unpleasant to solve by hand. Strict definiteness can be checked with determinants instead. For a 2×2 matrix $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$, and a larger one is expanded along a row, as in the example below; the determinant equals the product of the eigenvalues, so $\det A < 0$ already makes A indefinite.

Definition: Leading principal minor. The **leading principal minor** Δ_k of an $n \times n$ matrix A is the determinant of the submatrix formed by the first k rows and the first k columns of A , $k = 1, \dots, n$. They are usually called just the **leading minors**.

Example. $A = \begin{pmatrix} 2 & 2 & 0 \\ 2 & 4 & -1 \\ 0 & -1 & 6 \end{pmatrix}$: $\Delta_1 = 2$, $\Delta_2 = 8 - 4 = 4$ and $\Delta_3 = \det A = 2 \cdot 23 - 2 \cdot 12 + 0 = 46 - 24 = 22$ (expansion along the first row). All three minors are positive, so the criterion below gives $A \succ 0$.

Theorem: Sylvester's criterion. Let A be a symmetric $n \times n$ matrix. Then

$$A \succ 0 \iff \Delta_k > 0 \text{ for all } k, \quad A \prec 0 \iff (-1)^k \Delta_k > 0 \text{ for all } k.$$

The second pattern means alternating signs starting with minus: $\Delta_1 < 0$, $\Delta_2 > 0$, $\Delta_3 < 0$ and so on.

Why it works, for 2×2 and $A_{11} \neq 0$: complete the square,

$$A_{11}h_1^2 + 2A_{12}h_1h_2 + A_{22}h_2^2 = A_{11}\left(h_1 + \frac{A_{12}}{A_{11}}h_2\right)^2 + \frac{\Delta_2}{\Delta_1}h_2^2.$$

If $\Delta_1 = A_{11} > 0$ and $\Delta_2 > 0$, both coefficients are positive, and the sum vanishes only when $h_2 = 0$ and then $h_1 = 0$. If $\Delta_1 < 0$ and $\Delta_2 > 0$, both coefficients are negative. In general Gaussian elimination does the same: it completes the squares one variable at a time, and the leading minors control the signs of the coefficients that appear.

6.4 What Sylvester's criterion cannot do

Trap. Sylvester's criterion speaks only about strict definiteness. If neither pattern holds, do not conclude that the matrix is indefinite: it may be semidefinite. $A = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ have the same leading minors, $(\Delta_1, \Delta_2) = (0, 0)$, but $A \preceq 0$ and $B \succeq 0$. The same minors give opposite answers, so the leading minors cannot decide semidefiniteness at all.

A correct test of semidefiniteness uses **all** principal minors: $A \succeq 0$ exactly when the determinants of all submatrices on a set of rows and the same set of columns are nonnegative. A 3×3 matrix has 3 leading but 7 principal minors; for the matrix A above they are $(2, 4, 6, 4, 12, 23, 22)$. For $\text{diag}(0, -1)$ the principal minors are $(0, -1, 0)$, and the negative one shows that it is not $\succeq 0$.

A practical rule

On paper, use Sylvester's criterion only to decide strict definiteness. If some $\Delta_k = 0$, compute the eigenvalues instead; for a 2×2 matrix this is one quadratic equation. In code, always compute the eigenvalues and compare them with zero using a tolerance, because floating-point arithmetic rarely produces an exact zero (see the section "From the board to the keyboard").

Check yourself. Check $A = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$ with Sylvester's criterion. What can you conclude?

Answer. $(\Delta_1, \Delta_2) = (2, -4)$. The pattern of $A \succ 0$ needs every $\Delta_k > 0$ and fails at Δ_2 ; the pattern of $A \prec 0$ needs $\Delta_1 < 0$ and fails at once. Neither pattern holds, so the criterion only says that A is not definite. The eigenvalues $(-2, 2)$ have opposite signs and show that the matrix is indefinite.

7 Optimality conditions

Taylor's formula, definiteness and the difference between interior and boundary points now combine into four statements: a lemma on the directional derivative and three optimality conditions. Together they are the core of the first lecture of the course.

7.1 Feasible directions

From an interior point of Ω you can move a little in any direction and stay in Ω . From a boundary point you cannot: some directions leave Ω at once.

Definition: Feasible direction. A vector $s \in \mathbb{R}^n$, $s \neq 0$, is a **feasible direction** at a point $x \in \Omega$ if there is $\alpha_0 > 0$ such that $x + \alpha s \in \Omega$ for all $\alpha \in [0, \alpha_0]$.

In words: a small enough step along s keeps the point feasible. The number α_0 may be tiny, but it must be positive.

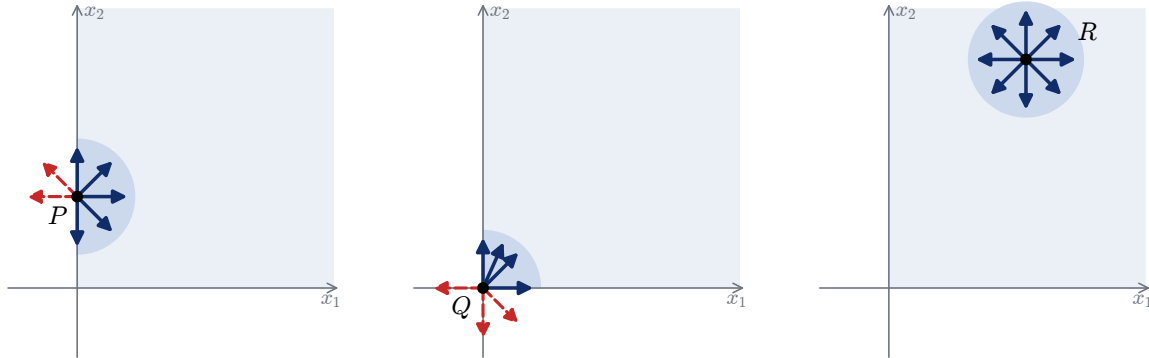


Figure. Feasible directions in the quadrant $\Omega = \{x_1 \geq 0, x_2 \geq 0\}$. At the boundary point $P = (0, 2)$ they form the half-plane $s_1 \geq 0$ (problem D1), at the corner $Q = (0, 0)$ the quarter $s_1 \geq 0, s_2 \geq 0$ (home practice H3) and at the interior point $R = (3, 5)$ every direction is feasible. The dashed red directions leave Ω at once.

If s is feasible, so is ts for every $t > 0$: the feasible directions form a **cone**. For a convex Ω they form, together with 0, a convex cone; for a nonconvex Ω they need not. At the origin of $\{x_1 \leq 0\} \cup \{x_2 \leq 0\}$ the feasible directions are $\{s \neq 0 : s_1 \leq 0 \text{ or } s_2 \leq 0\}$, and this set is not convex even with 0 added.

7.2 The directional derivative

Definition: Directional derivative. Let $x \in \Omega$ and let s be a direction. The **directional derivative** of f at x along s is the one-sided limit

$$\frac{\partial f}{\partial s}(x) = \lim_{\alpha \rightarrow 0^+} \frac{f(x + \alpha s) - f(x)}{\alpha}.$$

This is the ordinary derivative from a first calculus course, with the step taken along the vector s instead of along an axis. The limit is one-sided because at a boundary point only steps with $\alpha \geq 0$ may be allowed. At an interior point of a differentiable f both one-sided limits exist and agree, so the two-sided limit is not a mistake there.

Lemma: The directional derivative through the gradient. If f is differentiable at x , then $\frac{\partial f}{\partial s}(x) = s^\top \nabla f(x)$ for every direction s .

Proof. Put $h = \alpha s$ into Taylor's formula:

$$f(x + \alpha s) = f(x) + \alpha s^\top \nabla f(x) + o(\alpha).$$

Subtract $f(x)$, divide by $\alpha > 0$ and let $\alpha \rightarrow 0^+$: the term $o(\alpha)/\alpha$ tends to 0, and $s^\top \nabla f(x)$ remains.

Why this lemma matters

A limit, which takes work to compute, becomes an inner product, which takes one line. The lemma also explains why the gradient points in the direction of steepest ascent: among directions of length 1, the rate $s^\top \nabla f$ is largest for $s = \nabla f / \|\nabla f\|$ (when $\nabla f \neq 0$), where it equals $\|\nabla f\|$, because $s^\top \nabla f \leq \|s\| \|\nabla f\|$ by the Cauchy–Schwarz inequality.

Example. $f = x_1^2 + 2x_2^2$, $x = (1, 1)$, $s = (1, -1)$. The gradient is $\nabla f = (2x_1, 4x_2)^\top$, at x it equals $(2, 4)$, so $s^\top \nabla f(x) = -2$: f decreases along s . The fastest decrease per unit length is along $-\nabla f(x) / \|\nabla f(x)\|$, with rate $-\|\nabla f(x)\| = -2\sqrt{5} \approx -4.4721$, about 3.1623 times the rate along $s/\|s\|$ (see the trap below). This is why gradient descent steps against the gradient.

Trap. The number -2 is a rate per unit of α , not per unit of length. Here $\|s\| = \sqrt{2} \approx 1.4142$, so per unit length the rate is $-2/\|s\| = -\sqrt{2} \approx -1.4142$. Compare directions only after normalizing them; otherwise the longest vector looks the steepest.

7.3 The first-order necessary condition

Theorem: First-order necessary condition. Let $f \in C^1$ on Ω and let x^* be a local minimum of f on Ω . Then

$$s^\top \nabla f(x^*) \geq 0$$

for every feasible direction s at x^* . If, in addition, x^* is an interior point of Ω , then $\nabla f(x^*) = 0$.

Why. If f decreased along a direction in which you are allowed to move, a small step in that direction would give a feasible point with a smaller value, and x^* would not be a minimum.

Proof. Suppose that s is feasible and $s^\top \nabla f(x^*) = -\delta < 0$. By Taylor’s formula

$$f(x^* + \alpha s) = f(x^*) - \alpha \delta + o(\alpha),$$

and $|o(\alpha)| < \frac{1}{2}\alpha\delta$ for all small $\alpha > 0$. For these α

$$f(x^* + \alpha s) < f(x^*) - \alpha \delta + \frac{1}{2}\alpha\delta = f(x^*) - \frac{1}{2}\alpha\delta < f(x^*).$$

For small α the point $x^* + \alpha s$ lies in Ω , because s is feasible, and is as close to x^* as needed: a contradiction with the local minimality of x^* . Hence $s^\top \nabla f(x^*) \geq 0$. At an interior point both s and $-s$ are feasible for every $s \neq 0$, so $s^\top \nabla f(x^*) \geq 0$ and $-s^\top \nabla f(x^*) \geq 0$, that is $s^\top \nabla f(x^*) = 0$ for all s . The only vector orthogonal to every vector is zero, so $\nabla f(x^*) = 0$.

In plain words

From a minimum you cannot go downhill in any direction in which you are allowed to move. The familiar condition $\nabla f = 0$ is not a separate theorem but the special case of an interior point: there you may move both ways along every line, and “not downhill both ways” means “flat”. In Seminar 2 this inequality becomes the KKT conditions.

Definition: Stationary point. A point x with $\nabla f(x) = 0$ is a **stationary point** of f . Stationary points are the candidates for an unconstrained minimum: by the theorem above, a local minimum at an interior point is always stationary.

7.4 The second-order necessary condition

Theorem: Second-order necessary condition. Let $f \in C^2$ on Ω , let x^* be a local minimum of f on Ω and let s be a feasible direction at x^* with $s^\top \nabla f(x^*) = 0$. Then $s^\top H(x^*) s \geq 0$. In particular, at an interior point $\nabla f(x^*) = 0$ and $H(x^*) \succeq 0$.

Why. When the linear term of Taylor's formula vanishes along s , the quadratic term takes over, so the curvature along s decides. Negative curvature along s would mean that f goes down along s .

Proof. Put $h = \alpha s$ into Taylor's formula; the linear term vanishes because $s^\top \nabla f(x^*) = 0$:

$$f(x^* + \alpha s) = f(x^*) + \frac{1}{2} \alpha^2 s^\top H s + o(\alpha^2), \quad H = H(x^*).$$

If $s^\top H s = -\delta < 0$, then $|o(\alpha^2)| < \frac{1}{4} \alpha^2 \delta$ for all small $\alpha > 0$, and for these α

$$f(x^* + \alpha s) < f(x^*) - \frac{1}{2} \alpha^2 \delta + \frac{1}{4} \alpha^2 \delta = f(x^*) - \frac{1}{4} \alpha^2 \delta < f(x^*).$$

As in the previous proof, the point $x^* + \alpha s \in \Omega$ lies arbitrarily close to x^* , which contradicts the local minimality of x^* .

7.5 The second-order sufficient condition

Theorem: Second-order sufficient condition. Let $f \in C^2$ and let x^* be an **interior** point of Ω with $\nabla f(x^*) = 0$ and $H(x^*) \succ 0$. Then x^* is a strict local minimum.

Why. A positive definite Hessian means positive curvature in all directions at once, with a margin $\mu = \lambda_{\min} > 0$, and this margin beats the remainder of Taylor's formula.

Proof. Let $H = H(x^*)$ and $\mu = \lambda_{\min}(H) > 0$. In the axes of the eigenvectors $h^\top H h = \lambda_1 u_1^2 + \dots + \lambda_n u_n^2 \geq \mu \|h\|^2$. Since $\nabla f(x^*) = 0$, Taylor's formula gives

$$f(x^* + h) - f(x^*) = \frac{1}{2} h^\top H h + o(\|h\|^2) \geq \frac{1}{2} \mu \|h\|^2 + o(\|h\|^2).$$

There is $\varepsilon > 0$ such that the ball $\|h\| < \varepsilon$ around x^* lies in Ω and $|o(\|h\|^2)| < \frac{1}{4} \mu \|h\|^2$ for $0 < \|h\| < \varepsilon$. For these h

$$f(x^* + h) - f(x^*) > \frac{1}{4} \mu \|h\|^2 > 0.$$

So every other point of this neighbourhood has a larger value: x^* is a strict local minimum.

7.6 Summary and procedure

Order	Necessary	Sufficient
first order	$\nabla f(x^*) = 0$ at an interior point; $s^\top \nabla f(x^*) \geq 0$ for every feasible s at a boundary point	none
second order	$\nabla f(x^*) = 0$ and $H(x^*) \succeq 0$ at an interior point	$\nabla f(x^*) = 0$ and $H(x^*) \succ 0$ at an interior point

There is no sufficient condition of first order. The gradient only measures the slope, and the slope is zero at a minimum, at a maximum and at a saddle alike: $x_1^2 + x_2^2$, $-x_1^2 - x_2^2$ and $x_1^2 - x_2^2$ all have

$\nabla f(0) = 0$. Telling them apart needs information about the curvature.

The procedure

First find the candidates, the points where $\nabla f = 0$, and compute H at each of them. If $H \succ 0$, the candidate is a strict local minimum; if $H \prec 0$, a strict local maximum; if H is indefinite, a saddle. If H is singular and semidefinite, the second-order test cannot decide, and f itself has to be analysed near the point. Finally, call a minimum global only with an extra argument: convexity or a bound from below.

7.7 The gap between semidefinite and definite

Between the necessary and the sufficient condition of second order there is a gap: $H \succeq 0$ but not $H \succ 0$, a semidefinite Hessian with a zero eigenvalue. It may seem that the theory is just weak here and that a sharper test could close the gap. It cannot: no criterion that uses only ∇f and H at the point can close it, and the next example proves this.

Same data, different answers

$f_1 = x_1^4 + x_2^4$ and $f_2 = x_1^3 - 3x_1x_2^2$ both have $\nabla f(0) = 0$ and $H(0) = 0$. Yet f_1 has a strict global minimum at the origin, while f_2 has no extremum there (problems B4 and B3). The data of first and second order are the same and the correct answers differ, so no test that sees only $\nabla f(0)$ and $H(0)$ can tell the two functions apart.

In such a case, return to the definition of a minimum and work with f itself. One of two techniques usually helps:

- **Bound from below.** $f_1 = x_1^4 + x_2^4 \geq 0 = f_1(0)$, with equality only at the origin, so the origin is a strict global minimum. One line, no derivatives.
- **Find a direction of decrease.** Restrict f_2 to the line $x_2 = 0$: $f_2(x_1, 0) = x_1^3$ changes sign at 0. So arbitrarily close to the origin f_2 takes both positive and negative values, and the origin is not an extremum.

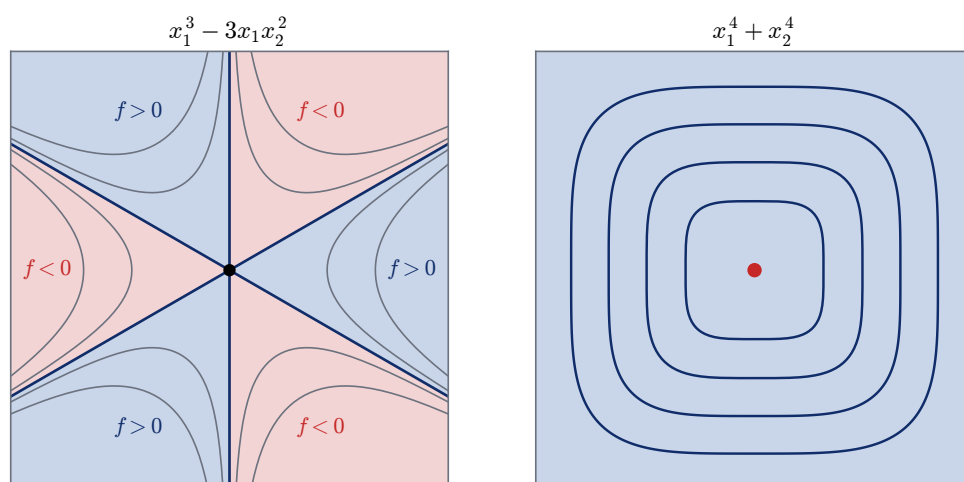


Figure. Left: the sign of $f_2 = x_1^3 - 3x_1x_2^2$ around the origin, 6 sectors of 60° where $f_2 > 0$ (blue) and $f_2 < 0$ (red) alternate. Right: $f_1 = x_1^4 + x_2^4$ is positive everywhere except the origin. Both functions have $\nabla f(0) = 0$ and $H(0) = 0$.

Check yourself. Let $g(x_1, x_2) = x_1^2$ on $\mathbb{R}^2$. Find its stationary points and its Hessian. What does the theory say about these points?

Answer. $\nabla g = (2x_1, 0)^\top$ vanishes on the whole line $x_1 = 0$, and $H = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} \succeq 0$ but not $\succ 0$, so the sufficient condition does not apply. Still, $g \geq 0$ and $g = 0$ on the line, so every point of the line is a non-strict global minimum: the flat bottom of a trough.

7.8 Why the monkey saddle has six sectors

The graph of $f_2 = x_1^3 - 3x_1x_2^2$ is the monkey saddle, and its picture can be explained completely. In polar coordinates $x_1 = r \cos \varphi$, $x_2 = r \sin \varphi$ the function becomes $f_2 = r^3 \cos^3 \varphi - 3r^3 \cos \varphi \sin^2 \varphi$. Replacing $\sin^2 \varphi$ by $1 - \cos^2 \varphi$ and using the triple-angle identity $\cos 3\varphi = 4\cos^3 \varphi - 3\cos \varphi$ gives

$$f_2 = r^3 (\cos^3 \varphi - 3\cos \varphi (1 - \cos^2 \varphi)) = r^3 (4\cos^3 \varphi - 3\cos \varphi) = r^3 \cos 3\varphi.$$

The factor r^3 is positive for $r > 0$, so the sign of f_2 depends only on the angle φ : going once around the circle, the cosine factor changes sign 6 times. This gives 6 sectors of 60° , 3 going up and 3 going down.

Why. An ordinary saddle has two descending sectors, one for each leg of a rider. This one has three: room for the two legs and the tail of a sitting monkey, hence the name.

Check yourself. How many sign sectors does $f = x_1^4 - 6x_1^2x_2^2 + x_2^4$ have around the origin, and what does the second-order test say there?

Answer. In polar coordinates $f = r^4 \cos 4\varphi$, so the sign changes 8 times around the circle: 8 sectors. The origin is the only stationary point and $H(0) = 0$, so the test gives no verdict; the sign changes show that the origin is not an extremum.

8 Convexity

Convexity is the one widely available property that turns local statements into global ones, which is why both theory and machine learning hold on to it.

8.1 Convex sets

Definition: Convex set. A set Ω is **convex** if together with any two of its points it contains the whole segment between them: $\lambda x + (1 - \lambda)y \in \Omega$ for all $x, y \in \Omega$ and $\lambda \in [0, 1]$.

The point $\lambda x + (1 - \lambda)y$ runs over the segment from y ($\lambda = 0$) to x ($\lambda = 1$); $\lambda = \frac{1}{2}$ gives the midpoint. A proof of convexity must show that the segment stays inside for every pair of points; to disprove convexity, one pair whose segment leaves the set is enough, as for the ring in the figure.

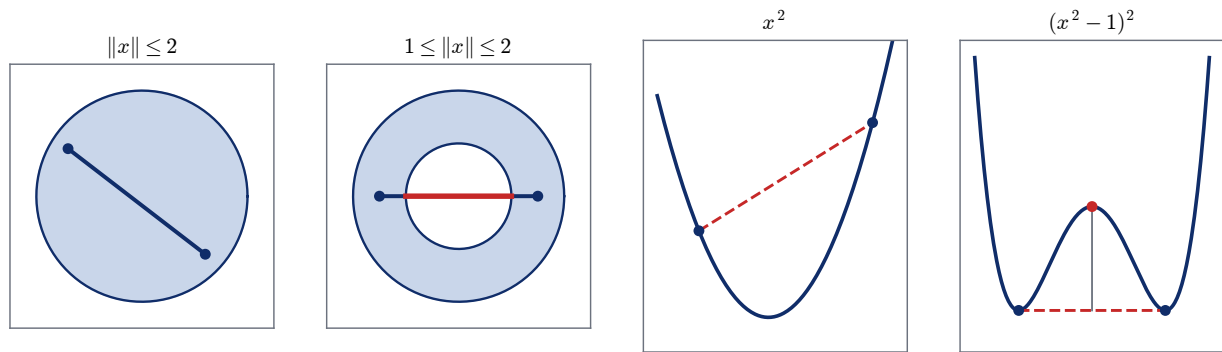


Figure. From left to right: the disk $\|x\| \leq 2$ is convex (problem C1); the ring $1 \leq \|x\| \leq 2$ is not, because the red part of the segment crosses the hole; x^2 is a convex function, its chord lies above the graph; $(x^2 - 1)^2$ is not convex, the chord between its two minima lies below the graph.

Example. Every ball $\{x : \|x - c\| \leq r\}$ in every norm is convex, including the ℓ_1 diamond and the ℓ_∞ square, which have corners. The proof in problem C1 uses only the triangle inequality and homogeneity of the norm: convexity is not smoothness.

8.2 Convex functions

Definition: Convex function. A function f on a convex set Ω is **convex** if

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$

for all $x, y \in \Omega$ and $\lambda \in [0, 1]$. It is **strictly convex** if the inequality is strict whenever $x \neq y$ and $0 < \lambda < 1$.

The left side is the value of the function at a point of the segment, the right side is the value on the chord that joins the endpoints of the graph: the graph never rises above a chord. In particular, f is convex along every line: for fixed x and s the function of one variable $t \mapsto f(x + ts)$ is convex where it is defined, so a single line along which this function is not convex shows that f is not convex.

Theorem: Second-order criterion. Let $f \in C^2$ on an open convex set Ω . Then f is convex $\iff H(x) \succeq 0$ for all $x \in \Omega$. If $H(x) \succ 0$ for all $x \in \Omega$, then f is strictly convex.

The converse of the last part fails: $f(x) = x^4$ is strictly convex, yet $f''(0) = 0$. It is the same gap between $\succeq$ and $\succ$ as in the optimality conditions.

Check yourself. Is $f = x_1^2 + x_2^2 + a x_1 x_2$ convex for $a = 3$?

Answer. No. The Hessian $H = \begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix}$ has the eigenvalues $2 \pm a$, that is $(-1, 5)$; one of them is negative, so f is not convex. It is convex exactly for $|a| \leq 2$ (problem C2).

8.3 Every local minimum is global

Theorem: Local minimum of a convex function. If f is convex on a convex set Ω and x^* is a local minimum, then x^* is a global minimum.

Why. A better point far away would pull the whole chord, and with it the graph, below $f(x^*)$, so points arbitrarily close to x^* would be better too.

Proof. Suppose $f(y) < f(x^*)$ for some $y \in \Omega$ and put $z_\lambda = \lambda y + (1 - \lambda)x^*$, which lies in Ω because Ω is convex. Convexity of f gives $f(z_\lambda) \leq \lambda f(y) + (1 - \lambda)f(x^*)$, that is $f(z_\lambda) \leq f(x^*) - \lambda(f(x^*) - f(y)) < f(x^*)$ for $\lambda \in (0, 1]$, while $\|z_\lambda - x^*\| = \lambda\|y - x^*\| \rightarrow 0$ as $\lambda \rightarrow 0$. So arbitrarily close to x^* there are points with a smaller value: a contradiction with local minimality.

Corollary. For a convex $f \in C^1$ on $\mathbb{R}^n$, the equality $\nabla f(x^*) = 0$ is not only necessary but also sufficient for a global minimum.

Proof. Let $\nabla f(x^*) = 0$ and suppose $f(y) < f(x^*)$ for some y . Put $s = y - x^*$ and $\varphi(\lambda) = f(x^* + \lambda s)$. By convexity $\varphi(\lambda) \leq (1 - \lambda)\varphi(0) + \lambda\varphi(1)$, so $\frac{\varphi(\lambda) - \varphi(0)}{\lambda} \leq \varphi(1) - \varphi(0) < 0$ for all $\lambda \in (0, 1]$. Letting $\lambda \rightarrow 0^+$ gives $s^\top \nabla f(x^*) \leq f(y) - f(x^*) < 0$, which contradicts $\nabla f(x^*) = 0$.

8.4 Why this matters in machine learning

The empirical risk with a regularization term, $L(w) = \frac{1}{m} \sum_i \ell(y_i, w^\top a_i) + \rho \|w\|^2$ with $\rho \geq 0$, is convex whenever the loss ℓ is convex in its second argument: a composition with the affine map $w \mapsto w^\top a_i$ and a nonnegative sum keep convexity. The squared loss, the logistic loss and the Huber loss are convex, so for linear models every point with zero gradient is a global minimizer. A neural network loses this property already with one hidden layer, which is why people speak about its loss landscape.

Example. The line fit from the section on optimization problems: $L(w) = \frac{1}{m} \|Xw - y\|^2$ with $m = 3$ and $X = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}$. Then $\nabla L(w) = \frac{2}{m} X^\top (Xw - y)$ and $H_L = \frac{2}{m} X^\top X = \begin{pmatrix} 2 & 2 \\ 2 & \frac{10}{3} \end{pmatrix}$, the same at every w . Its entry H_{11} is positive and $\det H_L = \frac{8}{3} > 0$, so $H_L \succ 0$ and L is strictly convex. The condition $\nabla L = 0$ is the **normal equation** $X^\top Xw = X^\top y$, here $\begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix} w = (7, 10)$, and its solution $w^* = (\frac{5}{6}, \frac{3}{2})$ is the unique global minimizer, $L(w^*) = \frac{1}{18}$. With collinear features $X^\top X$ is singular, H_L is only semidefinite and the minimizer is no longer unique; the regularization term $\rho \|w\|^2$ adds $2\rho I$ to H_L .

Why. For a convex loss, a method that drives the gradient to zero has found the answer; for a nonconvex loss it has found a candidate.

9 From the board to the keyboard

Every statement of the seminar has a short numerical counterpart. Two functions are enough: an independent check of a gradient formula and a second-order test with a tolerance; they are the starting point of home practice H4. The notebook of this seminar checks every number of this handout by two independent computations.

9.1 Checking a gradient: central differences

```
def central_diff(f, x, h=1e-5):
    """Gradient by central differences: an independent check of an analytic formula."""
    x = np.asarray(x, dtype=float)
    g = np.empty_like(x)
    for j in range(x.size):
        e = np.zeros_like(x)
        e[j] = h
        g[j] = (f(x + e) - f(x - e)) / (2 * h)
    return g
```

`central_diff` approximates each partial derivative by a central difference,

$$\frac{\partial f}{\partial x_j}(x) \approx \frac{f(x + he_j) - f(x - he_j)}{2h},$$

with an error of order h^2 ; here e_j is the j -th coordinate vector.

- **The check must be independent:** if the difference reuses parts of the analytic gradient, it checks itself.
- **The step has an optimum:** the truncation error $\sim h^2$ competes with the rounding error $\sim \varepsilon_{\text{mach}}/h$, so the best step is $h \sim \varepsilon_{\text{mach}}^{1/3} \approx 6 \cdot 10^{-6}$ with $\varepsilon_{\text{mach}} \approx 2.2 \cdot 10^{-16}$, hence the default $h = 10^{-5}$.
- **Symmetrize** a numerical Hessian before `eigvalsh`, which reads only one triangle of the matrix.

Example. For the lecture function at (1,1), `central_diff` returns (11,12) up to an error below 10^{-8} , in agreement with the formula for ∇f .

9.2 Classifying a Hessian with a tolerance

```
def classify(H, rel_tol=1e-9):
    """Second-order test for a symmetric matrix: "min", "max", "saddle" or "
    inconclusive".

    It reads the matrix only, so call it at a stationary point. The tolerance is
    relative to the largest |eigenvalue|.
    """
    H = np.asarray(H, dtype=float)
    w = np.linalg.eigvalsh(0.5 * (H + H.T))
    tol = rel_tol * np.abs(w).max(initial=0.0)
    if np.all(w > tol):
        return "min"
    if np.all(w < -tol):
        return "max"
    if w.min() < -tol and w.max() > tol:
        return "saddle"
    return "inconclusive"
```

`classify` reads the signs of the eigenvalues of the symmetrized matrix with a tolerance relative to the largest absolute eigenvalue, so rescaling the matrix does not change the answer. A matrix that is zero up to rounding has no scale: build Hessians exactly, by hand or with `sympy`, before calling it; a finite-difference Hessian of $f_1 = x_1^4 + x_2^4$ at the origin is classified as "min".

Matrix	Where it comes from	Output
$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$	problem B1 at the origin	"min"
$\begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$	problem B2 at the origin	"saddle"
$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$	problems B3 and B4 at the origin	"inconclusive"
$\begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}$	home practice H1 at the origin	"inconclusive"
$\begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}$	the Sylvester trap	"inconclusive"
$\begin{pmatrix} 36 & -4 \\ -4 & 16 \end{pmatrix}$	problem A2 at $(2, -1)$	"min"

Trap. `classify` looks at a matrix, not at a point. For the function of problem A2 at $(2, -1)$ it returns "min", because $H = \begin{pmatrix} 36 & -4 \\ -4 & 16 \end{pmatrix} \succ 0$, yet $\nabla f = (38, -16) \neq 0$: the point is not even stationary. Check the gradient first.

Trap. Do not write the saddle test as the chained comparison `w.min() < -tol < w.max()`. It means `w.min() < -tol` and `-tol < w.max()`, which is also true when the largest eigenvalue is 0: the matrix $\text{diag}(0, -1)$, which is negative semidefinite, would be called a saddle. A saddle needs `w.min() < -tol` and `w.max() > tol`.

The two "inconclusive" answers for B3 and B4 are not a weakness of the code: from ∇f and H alone the mathematics cannot decide (see the subsection on the gap between semidefinite and definite).

10 Questions students ask

Why do we need necessary conditions if they prove nothing?

They turn an infinite search into a finite one: instead of checking every feasible point, solve $\nabla f = 0$ and check the candidates. A point that violates a necessary condition is ruled out at once. Numerical methods follow the same idea: every gradient-based method of the first part of the course drives ∇f towards zero.

Why not just look at the graph?

For $n \leq 2$ you can, and the pictures of this handout do. For a linear model with 50 weights or a network with about 10^9 parameters there is no graph to look at, while the formulas work in any dimension. Even in the plane a picture can mislead: without level values a pit and a hill look the same. A picture gives intuition; a proof is a bound or a restriction to a line.

Is the Hessian always symmetric?

For the functions of this course, yes: by Schwarz's theorem the mixed second derivatives are equal when they are continuous, and counterexamples need discontinuous second derivatives. In code,

symmetrize anyway: in a numerical Hessian the entries H_{ij} and H_{ji} differ in the last digits, and `eigvalsh` reads only one triangle.

What does $o(\alpha)$ mean, and can we do without it?

It is the precise way to say “an error negligible for small α ”, precise enough to prove theorems with: $g(\alpha) = o(\alpha)$ means $g(\alpha)/\alpha \rightarrow 0$ as $\alpha \rightarrow 0$. For example, $f(x+h) = f(x) + \nabla f(x)^\top h + o(\|h\|)$ states that the error of the linear approximation vanishes faster than $\|h\|$. Without it every statement would be approximate.

11 Seminar problems

All eleven problems with complete solutions, grouped by the four blocks of the seminar. In each block the first problem is worked through in class and the others are for solving on your own.

11.1 A. Gradients and Hessians

A1. Gradient and Hessian of a quadratic

Partial derivatives term by term; a cross term in both off-diagonal entries

Statement. Let $f(x) = x_1^2 + x_2^2 - x_1x_2$. **(a)** Before computing: will the Hessian H depend on the point? **(b)** Find ∇f and H at an arbitrary point and at $(1, 1)$. **(c)** Is $(1, 1)$ a candidate for a minimum?

Solution.

1. Differentiate term by term. For x_1 : x_1^2 gives $2x_1$, x_2^2 gives nothing, $-x_1x_2$ gives $-x_2$. So $\partial f/\partial x_1 = 2x_1 - x_2$ and, in the same way, $\partial f/\partial x_2 = 2x_2 - x_1$.

2. Second derivatives:

$$H = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

No variable is left: the Hessian of a quadratic is constant. The cross term $-x_1x_2$ puts -1 into **both** off-diagonal entries.

3. At $(1, 1)$: $\nabla f = (1, 1) \neq 0$. The point is not stationary, so it is not a candidate for an unconstrained minimum, whatever H says.

4. The eigenvalues solve $\lambda^2 - 4\lambda + 3 = 0$: $\lambda = 1$ and $\lambda = 3$. Both are positive at every point, so f is strictly convex and its unique minimizer solves $\nabla f = 0$: $x^* = (0, 0)$.

5. Two conventions, one function: $f = \frac{1}{2}x^\top Hx$ with H itself, or $f = x^\top Ax$ with $A = \begin{pmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{pmatrix}$, whose off-diagonal entry is $-\frac{1}{2}$; $H = A + A^\top$. Mixing the two is what produces wrong Hessians.

Answer. $\nabla f = (2x_1 - x_2, 2x_2 - x_1)^\top$, $H = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ at every point; $\nabla f(1, 1) = (1, 1) \neq 0$, so $(1, 1)$ is not a candidate.

A2. The Hessian moves with the point

Substitution with signs; curvature of a non-quadratic function is local

Statement. For the lecture function $f(x) = 3x_1^3 + 2x_1x_2^2 + 4x_2^2$ compute ∇f , H and $\det H$ at $(2, -1)$. Compare with $\det H = 200$ at $(1, 1)$. Is $(2, -1)$ a stationary point?

Solution.

1. Differentiate: $\partial f/\partial x_1 = 9x_1^2 + 2x_2^2$, $\partial f/\partial x_2 = 4x_1x_2 + 8x_2$ and

$$H(x) = \begin{pmatrix} 18x_1 & 4x_2 \\ 4x_2 & 4x_1 + 8 \end{pmatrix}.$$

2. Keep the numbers visible: $\partial f/\partial x_1 = 9 \cdot 2^2 + 2 \cdot (-1)^2 = 38$ and $\partial f/\partial x_2 = 4 \cdot 2 \cdot (-1) + 8 \cdot (-1) = -16$, a sum of two negative terms. $H(2, -1) = \begin{pmatrix} 36 & -4 \\ -4 & 16 \end{pmatrix}$: both off-diagonal entries are negative.
3. $\det H = 36 \cdot 16 - (-4)^2 = 560$, against 200 at $(1, 1)$. For a non-quadratic function the curvature is a local object.
4. No: $\nabla f(2, -1) = (38, -16) \neq 0$, so the point is not stationary. $H \succ 0$ ($\Delta_1 = 36 > 0$, $\Delta_2 = 560 > 0$) says nothing about optimality at a non-stationary point.

Answer. $\nabla f = (38, -16)$, $H = \begin{pmatrix} 36 & -4 \\ -4 & 16 \end{pmatrix}$, $\det H = 560$; the point is not stationary.

A3. Three variables and Sylvester's criterion

A three-by-three Hessian, leading minors, the minimizer of a convex quadratic

Statement. Let $f(x) = x_1^2 + 2x_2^2 + 3x_3^2 + 2x_1x_2 - x_2x_3$ on $\mathbb{R}^3$. **(a)** Find ∇f and H . **(b)** Decide by Sylvester's criterion whether $H \succ 0$. **(c)** What does this say about convexity? Where is the minimum, and is it unique?

Solution.

1. The diagonal holds twice the coefficients of the squares, $(2, 4, 6)$. The term $2x_1x_2$ gives $H_{12} = H_{21} = 2$, the term $-x_2x_3$ gives $H_{23} = H_{32} = -1$, and x_1, x_3 never meet, so $H_{13} = 0$:

$$\nabla f = \begin{pmatrix} 2x_1 + 2x_2 \\ 2x_1 + 4x_2 - x_3 \\ 6x_3 - x_2 \end{pmatrix}, \quad H = \begin{pmatrix} 2 & 2 & 0 \\ 2 & 4 & -1 \\ 0 & -1 & 6 \end{pmatrix}.$$

2. $\Delta_1 = 2$, $\Delta_2 = 2 \cdot 4 - 2 \cdot 2 = 4$ and, expanding along the first row, $\Delta_3 = 2(4 \cdot 6 - 1) - 2(2 \cdot 6 - 0) + 0 = 46 - 24 = 22$.
3. All leading minors are positive, so $H \succ 0$. The Hessian is the same at every point, so f is strictly convex on $\mathbb{R}^3$.
4. Solve $\nabla f = 0$: $x_1 = -x_2$, $x_2 = 6x_3$, $x_3 = 2x_2$, hence $x_2 = 6x_3 = 12x_2$, so $x_2 = 0$ and then $x_3 = x_1 = 0$. The unique stationary point $x^* = (0, 0, 0)$ is the global minimizer by convexity.

Answer. $H = \begin{pmatrix} 2 & 2 & 0 \\ 2 & 4 & -1 \\ 0 & -1 & 6 \end{pmatrix}$, $\Delta = (2, 4, 22)$, so $H \succ 0$ and f is strictly convex; the unique global minimizer is $x^* = (0, 0, 0)$.

11.2 B. Classifying stationary points

B1. A bowl: local first, global separately

The full routine: stationary point, Hessian, sufficient condition, a separate global argument

Statement. Let $f(x) = x_1^2 + x_2^2$. **(a)** Find all stationary points. **(b)** Compute H and decide its definiteness. **(c)** Which condition did you use: necessary or sufficient? **(d)** Is the point a global minimum, and what exactly justifies that?

Solution.

1. $\nabla f = (2x_1, 2x_2)^\top = 0$ only at $x^* = (0, 0)$.
2. $H = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ with $\Delta_1 = 2 > 0$ and $\Delta_2 = 4 > 0$, so $H \succ 0$.
3. $\nabla f(x^*) = 0$ and $H(x^*) \succ 0$: the **sufficient** condition applies, and it gives a **strict local** minimum, nothing more.
4. Global needs one more sentence: either convexity ($H \succeq 0$ at every point) or directly $f(x) = x_1^2 + x_2^2 \geq 0 = f(x^*)$, with equality only at x^* .
5. The eigenvalues 2 and 2 are equal: the curvature is the same in every direction and the level sets are circles. Gradient descent with the step size $\frac{1}{2}$ reaches x^* in one step from any x : $x - \frac{1}{2}\nabla f(x) = 0$.

Answer. The only stationary point is $x^* = (0, 0)$; $H \succ 0$ gives a strict local minimum by the sufficient condition; it is global because $f \geq 0 = f(x^*)$ (or by convexity).

B2. A saddle, proved without a picture

Indefinite Hessian; restriction to two lines as the proof

Statement. Let $f(x) = x_1^2 - x_2^2$. Show that $(0, 0)$ is a stationary point that is neither a minimum nor a maximum, without using a picture.

Solution.

1. $\nabla f = (2x_1, -2x_2)^\top$ vanishes only at the origin, and $H = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$.
2. $\Delta_1 = 2 > 0$ and $\Delta_2 = -4 < 0$: neither. Since $\det H < 0$, the eigenvalues -2 and 2 have opposite signs, and H is indefinite.
3. The proof restricts f to the two axes: $f(t, 0) = t^2 > 0$ and $f(0, t) = -t^2 < 0$ for $t \neq 0$. Every neighbourhood of the origin contains larger and smaller values than $f(0, 0) = 0$: a saddle.
4. Along the line at angle θ : $f(t \cos \theta, t \sin \theta) = t^2 \cos 2\theta$, which vanishes identically at $\theta = 45^\circ, 135^\circ$ and has opposite signs on the two sides of these lines. The picture is a guide; the two restrictions are the proof.

Answer. $\Delta_1 = 2$, $\Delta_2 = -4$, H indefinite; $f(t, 0) = t^2$ and $f(0, t) = -t^2$, so the origin is a saddle.

B3. The monkey saddle

A zero Hessian: every optimality condition is silent, the definition decides

Statement. Let $f(x) = x_1^3 - 3x_1x_2^2$. **(a)** Find all stationary points. **(b)** Evaluate the Hessian at the origin. **(c)** Try the optimality conditions: what do they say? **(d)** Find a direction that settles the question.

Solution.

1. $\partial f/\partial x_1 = 3x_1^2 - 3x_2^2$, $\partial f/\partial x_2 = -6x_1x_2$. Case split: $-6x_1x_2 = 0$ gives $x_1 = 0$ **or** $x_2 = 0$, and in either case the first equation forces the other coordinate to vanish. The only stationary point is the origin.
2. $H(x) = \begin{pmatrix} 6x_1 & -6x_2 \\ -6x_2 & -6x_1 \end{pmatrix}$, so $H(0,0) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$, the zero matrix.
3. The zero matrix is $\succeq 0$, so the necessary condition for a minimum holds; it is also $\preceq 0$, so the one for a maximum holds too. The sufficient condition needs $H \succ 0$ and does not apply, and all $\Delta_k = 0$, so Sylvester is silent. Every tool is silent.
4. Use the definition along a line: $f(t,0) = t^3$ is positive for $t > 0$ and negative for $t < 0$. Every neighbourhood of the origin has larger and smaller values: neither a minimum nor a maximum.
5. Why “monkey”: in polar coordinates $f = r^3 \cos 3\varphi$, so the sign alternates in six sectors of 60° , three up and three down: two valleys for the legs and one for the tail. Along $\theta = 30^\circ, 90^\circ, 150^\circ$ the function is identically zero, and those lines settle nothing.

Answer. The only stationary point is the origin; $H(0) = 0$, so the second-order test gives no verdict; $f(t,0) = t^3$ changes sign, so the origin is neither a minimum nor a maximum.

B4. Same data, opposite answer

A direct bound decides where the second-order test cannot

Statement. Let $f(x) = x_1^4 + x_2^4$. **(a)** Find the stationary points and $H(0,0)$. **(b)** What does the second-order test say? **(c)** Decide the question in one line. **(d)** Compare with B3: what is identical and what is different?

Solution.

1. $\nabla f = (4x_1^3, 4x_2^3)^\top = 0$ only at the origin, and $H(x) = \begin{pmatrix} 12x_1^2 & 0 \\ 0 & 12x_2^2 \end{pmatrix}$, so $H(0,0) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ again.
2. The second-order test is silent again: the necessary conditions hold and the sufficient one does not apply.
3. One line: $f(x) = x_1^4 + x_2^4 \geq 0 = f(0,0)$, with equality only at the origin. A **strict global** minimum, no theorem needed.
4. B3 and B4 have identical $\nabla f(0)$ and $H(0)$ and opposite answers. When $H(0)$ is singular and semidefinite, the second-order test provably cannot decide: two functions with the same first- and second-order data behave differently.

Answer. $H(0,0) = 0$ and the test is silent; $f \geq 0 = f(0,0)$ with equality only at the origin, so it is a strict global minimum. B3 has the same $\nabla f(0)$ and $H(0)$ and no extremum.

11.3 C. Convexity

C1. A disk is convex: a proof from the definition

A convexity proof with a named reason on every line

Statement. Let $\Omega = \{x \in \mathbb{R}^2 : x_1^2 + x_2^2 \leq 4\}$. **(a)** Prove from the definition that Ω is convex. **(b)** Which properties of the norm did the proof use? **(c)** Does the argument work for a square? For a ring?

Solution.

1. Rewrite the set as a ball: $\Omega = \{x : \|x\| \leq 2\}$. Take $x, y \in \Omega$, $\lambda \in [0, 1]$ and $z = \lambda x + (1 - \lambda)y$.

2.

$$\|z\| \stackrel{(1)}{\leq} \|\lambda x\| + \|(1 - \lambda)y\| \stackrel{(2)}{=} \lambda\|x\| + (1 - \lambda)\|y\| \leq 2\lambda + 2(1 - \lambda) = 2,$$

where (1) is the triangle inequality and (2) is homogeneity; λ and $1 - \lambda$ are nonnegative, so no absolute values appear. Hence $z \in \Omega$. ■

3. The proof used only the triangle inequality and homogeneity: nothing about circles, the dimension or smoothness. So every ball in every norm is convex, including the ℓ_1 diamond and the ℓ_∞ square, which have corners.

4. The ring $\{1 \leq \|x\| \leq 2\}$ is not convex: the points $(-1.5, 0)$ and $(1.5, 0)$ lie in it, but their midpoint $(0, 0)$ lies in the hole.

Answer. Convex, by the triangle inequality and homogeneity; the same proof covers the ℓ_∞ square, while the ring is not convex.

C2. It is the size of the cross term that matters

Convexity of a quadratic from one constant matrix; a one-parameter family

Statement. **(a)** Is $f(x) = x_1^2 + x_2^2 - 4x_1x_2$ convex on $\mathbb{R}^2$? **(b)** Compare with A1, where the cross coefficient was -1 . **(c)** For which a is $f_a(x) = x_1^2 + x_2^2 + ax_1x_2$ convex, and for which is it strictly convex?

Solution.

1. $H = \begin{pmatrix} 2 & -4 \\ -4 & 2 \end{pmatrix}$ at every point; $\Delta_1 = 2 > 0$ and $\Delta_2 = 2 \cdot 2 - (-4)^2 = -12 < 0$.

2. $\Delta_2 = \det H < 0$, so the eigenvalues -2 and 6 have opposite signs and H is indefinite (that neither Sylvester pattern holds would not be enough). H is nowhere $\succeq 0$, so f is **not** convex.

3. The family: $H_a = \begin{pmatrix} 2 & a \\ a & 2 \end{pmatrix}$ with eigenvalues $2 \pm a$. Convex $\iff$ both are $\geq 0 \iff |a| \leq 2$; strictly convex $\iff |a| < 2$ — for a quadratic the converse of the second-order criterion does hold, and the next step shows what happens at $|a| = 2$.

4. $f = (x_1 + x_2)^2$: convex but not strictly, the eigenvalues are 0 and 4 , and the whole line $x_1 + x_2 = 0$ consists of minimizers. A1 has $a = -1$ (eigenvalues $1, 3$, convex); here $a = -4$: the size of the cross term decides, not its sign.

Answer. $\Delta_2 = -12$, eigenvalues -2 and 6 : not convex. f_a is convex for $|a| \leq 2$ and strictly convex for $|a| < 2$.

11.4 D. Feasible directions

D1. Optimality on the boundary

Active constraints, the set of feasible directions and one violating direction

Statement. Let $\Omega = \{x : x_1 \geq 0, x_2 \geq 0\}$, $f(x) = x_1 + x_2$ and $x = (0, 2)$. **(a)** Which constraints are active at x ? **(b)** Describe all feasible directions at x . **(c)** Check the first-order necessary condition $s^\top \nabla f(x) \geq 0$. **(d)** If it fails, give a direction that breaks it.

Solution.

1. $x_1 = 0$: the constraint $x_1 \geq 0$ is **active**. $x_2 = 2 > 0$: the constraint $x_2 \geq 0$ is inactive, with room on both sides.
2. The feasible directions are $\{s \neq 0 : s_1 \geq 0\}$, with s_2 free. If $s_1 < 0$, then $x_1 + \alpha s_1 < 0$ for every $\alpha > 0$. If $s_1 \geq 0$, the first coordinate stays nonnegative, and $x_2 + \alpha s_2 > 0$ for all $\alpha \in [0, \alpha_0]$, for example with $\alpha_0 = 1/|s_2|$ when $s_2 < 0$ (any $\alpha_0 > 0$ works when $s_2 \geq 0$).
3. $\nabla f \equiv (1, 1)$, so the condition to test is $s^\top \nabla f = s_1 + s_2 \geq 0$ for **every** feasible s .
4. Take $s = (0, -1)$: it is feasible, and $s^\top \nabla f = -1 < 0$. The condition fails, so $(0, 2)$ is not a candidate. Moving down the edge decreases f until the corner $(0, 0)$ with $f = 0$, the true minimizer.
5. To reject a point, one violating direction is enough; to accept it, every feasible direction must pass. Replacing this infinite check by finitely many multipliers is the idea of the KKT conditions in the next seminar.

Answer. Only $x_1 \geq 0$ is active; the feasible directions are $\{s \neq 0 : s_1 \geq 0\}$; $s = (0, -1)$ gives $s^\top \nabla f = -1 < 0$, so $(0, 2)$ is not a candidate.

D2. A directional derivative, two ways

The limit definition against $s^\top \nabla f$; rate per unit of α versus per unit of length

Statement. Let $f(x) = x_1^2 + 2x_2^2$, $x = (1, 1)$ and $s = (1, -1)$. **(a)** Compute $\partial f / \partial s(x)$ from the definition, as a limit. **(b)** Compute it again as $s^\top \nabla f(x)$. **(c)** Is s a descent direction? **(d)** Is your number a rate per unit of α or per unit of length?

Solution.

1. $x + \alpha s = (1 + \alpha, 1 - \alpha)$, so $f(x + \alpha s) = (1 + \alpha)^2 + 2(1 - \alpha)^2 = 3 - 2\alpha + 3\alpha^2$, and $f(x) = 3$ is its value at $\alpha = 0$.
- 2.

$$\frac{\partial f}{\partial s}(x) = \lim_{\alpha \rightarrow 0^+} \frac{f(x + \alpha s) - f(x)}{\alpha} = \lim_{\alpha \rightarrow 0^+} (-2 + 3\alpha) = -2.$$

The limit is one-sided because at a boundary point only $\alpha \geq 0$ may be allowed; at this interior point both one-sided limits agree.

3. $\nabla f = (2x_1, 4x_2)^\top$, so $\nabla f(x) = (2, 4)$ and $s^\top \nabla f(x) = 1 \cdot 2 + (-1) \cdot 4 = -2$: both routes agree.
4. The value is negative, so s is a descent direction. But $\|s\| = \sqrt{2}$, so -2 is a rate per unit of α ; per unit of length it is $-\sqrt{2} \approx -1.4142$.
5. The steepest descent direction is $-\nabla f(x) / \|\nabla f(x)\|$, with the rate $-\|\nabla f(x)\| = -2\sqrt{5} \approx -4.4721$ per unit of length, $\sqrt{10} \approx 3.1623$ times the rate along $s/\|s\|$. This is why descent methods step against the gradient.

Answer. $f(x + \alpha s) = 3 - 2\alpha + 3\alpha^2$, both routes give -2 per unit of α ; per unit of length $-\sqrt{2}$; s is a descent direction.

12 Home practice

Four problems for home: three by hand and one in code. Each is followed by its complete solution, so try it on your own first.

H1. A singular but nonzero Hessian

By hand

Statement. Find the stationary points of $f(x) = x_1^3 + x_2^2$ and classify them.

Solution.

1. $\nabla f = (3x_1^2, 2x_2)^\top = 0$ only at the origin.
2. $H(x) = \text{diag}(6x_1, 2)$, so $H(0) = \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}$ with eigenvalues 0 and 2: positive semidefinite, not definite. The necessary condition holds, the sufficient one does not apply.
3. Along $x_2 = 0$: $f(t, 0) = t^3$ changes sign, so the origin is neither a minimum nor a maximum. Compare with B3: there $H(0) = 0$; here $H(0)$ is not the zero matrix but singular, and the conclusion is the same.

Answer. The only stationary point is the origin; $H(0)$ has eigenvalues 0 and 2; neither a minimum nor a maximum.

H2. Minimizers of the quadratic family

By hand

Statement. Let $f_a(x) = x_1^2 + x_2^2 + a x_1 x_2$. **(a)** For $a = 5$ find the eigenvalues of H_a and the value $f_a(1, -1)$. **(b)** For which a does f_a have a unique minimizer? **(c)** Describe all minimizers at $a = 2$.

Solution.

1. The characteristic polynomial is $\lambda^2 - 4\lambda + 4 - a^2$, so the eigenvalues are $2 \pm a$. At $a = 5$ they are -3 and 7 , and $f_a(1, -1) = -3 < 0 = f_a(0)$. Since $f_a(tx) = t^2 f_a(x)$, along that direction f_a decreases without bound.
2. $|a| < 2$: $H_a \succ 0$, f_a is strictly convex and the origin is the unique minimizer. $|a| > 2$: H_a is indefinite and f_a is unbounded below, so there is no minimizer.
3. $a = 2$: $f = (x_1 + x_2)^2 \geq 0$, and the minimizers are the whole line $x_1 + x_2 = 0$ (for $a = -2$, the line $x_1 = x_2$).

Answer. At $a = 5$: eigenvalues -3 and 7 , $f_a(1, -1) = -3$. A unique minimizer exactly for $|a| < 2$; at $a = 2$ the minimizers form the line $x_1 + x_2 = 0$.

H3. Feasible directions at a corner

By hand

Statement. For $\Omega = \{x : x_1 \geq 0, x_2 \geq 0\}$ and $f(x) = x_1 + x_2$, describe the feasible directions at $x = (0, 0)$ and check the first-order necessary condition there.

Solution.

1. Both constraints are active, so the feasible directions are $\{s \neq 0 : s_1 \geq 0, s_2 \geq 0\}$.
2. $s^\top \nabla f = s_1 + s_2 > 0$ for every such s : the condition holds. Over unit feasible directions the smallest rate is 1, attained along the edges.
3. Indeed $f \geq 0$ on Ω and $f = 0$ at the corner, so the corner is the minimizer. Compare with D1, where one inactive constraint left room for the violating direction.

Answer. Both constraints are active; the feasible directions are $s \neq 0$ with $s_1 \geq 0$ and $s_2 \geq 0$; the condition holds, and the corner is the minimizer.

H4. The second-order test in code

In code

Statement. Using `classify` from block B, test the Hessians of today at their points: A1, A2 at $(2, -1)$, A3, B1, B2, B3 and B4 at the origin, C2, H1 at the origin and the pair `diag(0, -1)`, `diag(0, 1)`. Where does it answer "inconclusive", and why does A2 get "min"?

Solution.

1. Build each Hessian exactly (by hand or with `sympy`) as a `numpy` array and call `classify(H)`.
2. The labels: 4 times "min" (A1, A2, A3, B1), 2 times "saddle" (B2, C2) and 5 times "inconclusive" (B3, B4, H1 and both diagonal matrices).
3. None of the "inconclusive" answers is a bug: the second-order test genuinely cannot decide there. The version with the chained comparison `w.min() < -tol < w.max()` calls `diag(0, -1)` a saddle; the test must require a negative **and** a positive eigenvalue. A Hessian computed by finite differences is another matter: for B4 at the origin it is a tiny positive diagonal matrix, and `classify` answers "min"; no tolerance rule can tell an exact zero from a tiny positive number without a scale taken from the problem.
4. A2 gets "min" because `classify` reads the matrix only: $H(2, -1) \succ 0$, but the point is not stationary, so the label means nothing there.

Answer. "inconclusive" for B3, B4, H1, `diag(0, -1)` and `diag(0, 1)`; A2 gets "min" because the function reads H only and the point is not stationary.

13 Common mistakes

Mistakes that come up again and again: first by topic, then from the seminar problems.

13.1 Gradients and Hessians

- A cross term $c x_i x_j$ puts c into both H_{ij} and H_{ji} . Writing $c/2$ (the entry of A in $f = x^\top A x$) or doubling c as on the diagonal gives a wrong Hessian.
- Discussing the Hessian at a point before checking $\nabla f = 0$ there: at a non-stationary point $H \succ 0$ means nothing for optimality.
- Treating the Hessian of a cubic as constant: only a quadratic has the same H at every point.

From the problems:

- **A1.** Writing -2 off the diagonal (doubling the cross coefficient as on the diagonal) or $-\frac{1}{2}$ (the entry of A in $f = x^\top Ax$), or putting -1 into only one of the two entries.
- **A1.** Discussing optimality at $(1, 1)$ from the Hessian: the gradient is not zero there.
- **A2.** Losing a sign: at $x_2 < 0$ both off-diagonal entries are negative, and $\partial f / \partial x_2$ is a sum of two negative terms.
- **A2.** Announcing a minimum because $H \succ 0$ at a point where $\nabla f \neq 0$.
- **A3.** Halving the cross coefficients, or writing each of them into only one of the two symmetric entries.
- **A3.** Calling the stationary point a global minimum without an argument: here it is convexity, from $H \succ 0$ at every point.

13.2 Classifying stationary points

- Concluding “indefinite” because neither Sylvester pattern holds: $\text{diag}(0, -1)$ and $\text{diag}(0, 1)$ have the same leading minors, yet the first is negative semidefinite and the second positive semidefinite.
- Reading the necessary condition $H(x^*) \succeq 0$ as a certificate: $H = 0$ holds at the monkey saddle, which has no extremum.
- Claiming a global minimum from $\nabla f = 0$ and $H \succ 0$: the sufficient condition gives a strict local minimum only.
- Offering a contour picture as the proof of a saddle: restrictions to lines are the proof.

From the problems:

- **B1.** Claiming a global minimum from $H(x^*) \succ 0$ alone.
- **B2.** Offering the contour picture as the proof; it is only a guide.
- **B3.** Concluding a minimum from $H(0) = 0 \succeq 0$: that treats the necessary condition as if it were sufficient. This is the mistake the problem exists for.
- **B3.** Testing only the x_2 -axis, where $f \equiv 0$, and concluding that nothing happens.
- **B4.** Calling the silent test “bad luck”: with H singular and semidefinite, the first- and second-order data cannot separate B3 from B4.

13.3 Convexity

- Judging convexity of a quadratic by the sign of the cross term: its size decides.
- Confusing convexity with smoothness: the ℓ_1 diamond has corners and is convex, the ring has a smooth boundary and is not.
- Proving convexity with a drawing or with a few segments that stay inside: only a proof for all pairs counts.

From the problems:

- **C1.** Giving a drawing instead of a proof, or a chain of inequalities without the reason for each step.

- **C2.** Judging by the sign of the cross term: A1 and C2 both have a negative one.
- **C2.** Concluding “indefinite” only because neither Sylvester pattern holds: the negative determinant is what proves it here.

13.4 Feasible directions

- Declaring every direction feasible at a boundary point, or none.
- Accepting a boundary point after checking a few directions: rejecting needs one violating direction, accepting needs all of them.
- Comparing directional derivatives of directions with different lengths: normalize first, or the longest direction looks the steepest.

From the problems:

- **D1.** Declaring all directions feasible, or none.
- **D1.** Testing only directions that point into the interior and concluding that the condition holds.
- **D2.** Reporting -2 as the rate per unit of length: s is not a unit vector.
- **D2.** Forgetting to subtract $f(x)$ before dividing by α : the quotient $f(x + \alpha s)/\alpha$ has no finite limit.

14 Glossary

The English terms of the seminar with short meanings and their Russian equivalents.

Term	Meaning
optimization problem	finding the smallest value of a function over a set
objective function	the function being minimized; in machine learning, the loss
empirical risk	the average loss of a model over the training data
feasible set	the set of points among which the minimum is sought
interior point	a point that lies in the set together with a small ball around it
boundary point	a point of the set with points outside the set arbitrarily close to it
local minimum	a point with the smallest value among all nearby feasible points
global minimum	a point with the smallest value on the whole feasible set
strict minimum	a minimum where every other admissible point gives a strictly larger value
minimizer	the point at which a minimum is attained
stationary point	a point where the gradient is zero
saddle point	a stationary point that is neither a minimum nor a maximum
monkey saddle	a saddle with three descending sectors instead of two
level set	the set of points where a function takes a given value
partial derivative	the derivative with respect to one variable while the others are fixed
gradient	the column vector of all first partial derivatives
Hessian	the matrix of all second partial derivatives

Term	Meaning
directional derivative	the rate of change of a function when moving from a point along a direction
descent direction	a direction along which the function decreases to first order
steepest descent direction	the direction opposite to the gradient
Taylor's formula	the approximation of a smooth function near a point by a polynomial
quadratic form	the number $\mathbf{h}^T \mathbf{A} \mathbf{h}$ for a symmetric matrix $\mathbf{A}$ and a vector $\mathbf{h}$
cross term	a product of two different variables with a coefficient
eigenvalue	the stretching factor of a matrix along one of its eigenvectors
positive definite matrix	a symmetric matrix whose quadratic form is positive for every nonzero vector
positive semidefinite matrix	a symmetric matrix whose quadratic form is nonnegative for every vector
indefinite matrix	a symmetric matrix whose quadratic form takes both signs
determinant	the number $ad - bc$ of a two by two matrix, and the product of the eigenvalues in general
leading principal minor	the determinant of the upper-left square block of a matrix
principal minor	the determinant on a set of rows and the same set of columns
Sylvester's criterion	a symmetric matrix is positive definite exactly when all its leading principal minors are positive
necessary condition	a condition that every minimum satisfies; its failure rules a point out
sufficient condition	a condition that guarantees a minimum
feasible direction	a direction in which a small step keeps the point in the feasible set
active constraint	an inequality that holds with equality at the point
convex set	a set that contains the segment between any two of its points
convex function	a function whose graph lies on or below each of its chords
strictly convex function	a convex function for which the chord inequality is strict between distinct points
condition number	the ratio of the largest to the smallest eigenvalue of a positive definite matrix
central difference	an estimate of a derivative from the values of a function at two points symmetric around the given one
tolerance	the threshold below which a computed number is treated as zero

15 References

- [1] S. Boyd, L. Vandenberghe. *Convex Optimization*. Cambridge University Press. The chapters on convex sets and convex functions, and the appendix on mathematical background: norms, derivatives, second-order approximation and symmetric matrices.
- [2] Andrey Ignatov. Lectures of the course *Optimization Methods*, HSE University, Faculty of Computer Science: the lecture “Optimization as the Foundation of Machine Learning”.